

Machine Learning — Programming Assignment

June 2026

1 Problem definition and data

Despite the growing popularity of electronic payment systems, banknotes and coins remain crucial in many scenarios. Their automatic recognition is a widely explored task in computer vision.

A company asks you to design and implement a system for the automatic recognition of Euro coins. The classes correspond to the values 1, 2, 5, 10, 20, 50 cents, one Euro and two Euros. They collected a dataset with 100 training images and 20 test images of each kind. Images have been resized to 64×64 pixels.

The images have been taken with variable noise, sharpness, lighting conditions, and backgrounds. They can be face up or down, and can be rotated in any direction. Here are some examples:



2 Assignment

We want to build one or more classifiers that automatically recognize coin types.

For the programming assignment, you are expected to:

1. analyze and comment the data;
2. implement, train, and evaluate one or more classification models;

3. use suitable data processing and visualization techniques to analyze the behavior of the trained models.

All the above should be implemented as scripts in the Python programming language. The code written during the course can be used for the assignment if needed.

Special request: after a good model in terms of classification error has been found, the company want at least one additional model designed to minimize the “value of misclassification” measured as the mean absolute difference between the predicted and real face value of the coin (e.g., a 50-cent coin misclassified as a 20-cent coin would “cost” 30 points).

3 Report

Prepare a report of three to five pages documenting all your work. Provide detailed instructions on how to reproduce the results. The report must be in the PDF format. Include your name in the report and conclude the document with the following statement: “I affirm that this report is the result of my own work and that I did not share any part of it with anyone else except the teacher.”

Any use of AI tools in the solution of the assignment, or in the preparation of the report, must be declared by listing the tasks they have been used for.

Make a ZIP archive with the report and the Python notebook (converted to PDF, with visible outputs), and submit it from the course web page.